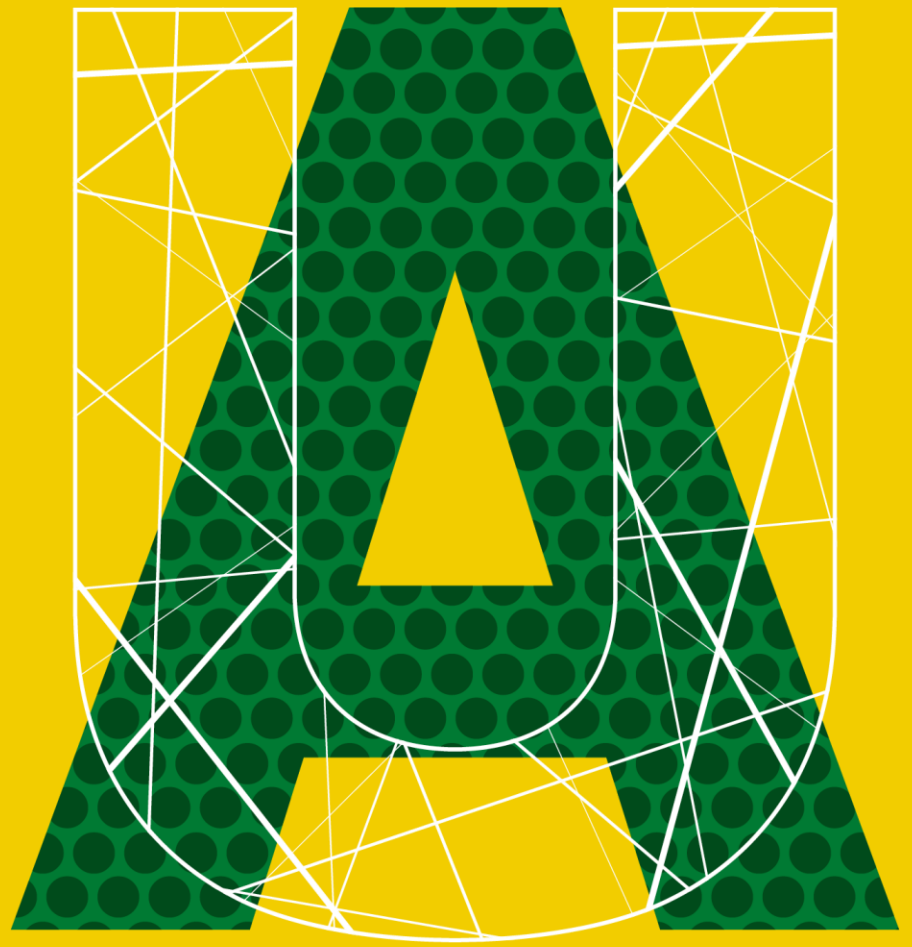
Efficient Robotic Pushing with Diffusion Policy

Adaptive Execution and Self-Imitation-Based Fine-Tuning

Qihang Feng

UNIVERSITY OF ALBERTA

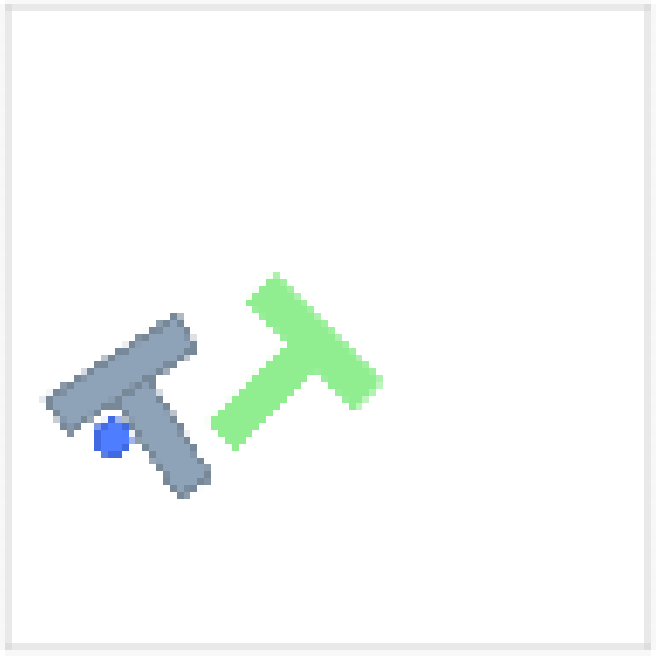


ECE 730 PROJECT MID-UPDATE

Computation and AI for Cyber-Physical Systems

PushT task and feedback loop

Push the T-block onto the green target



Recorded camera image

1. Observe

Two camera frames and pusher position

2. Predict

16 action positions from Diffusion Policy

3. Execute

8 or 12 movements with visual monitoring

4. Observe again

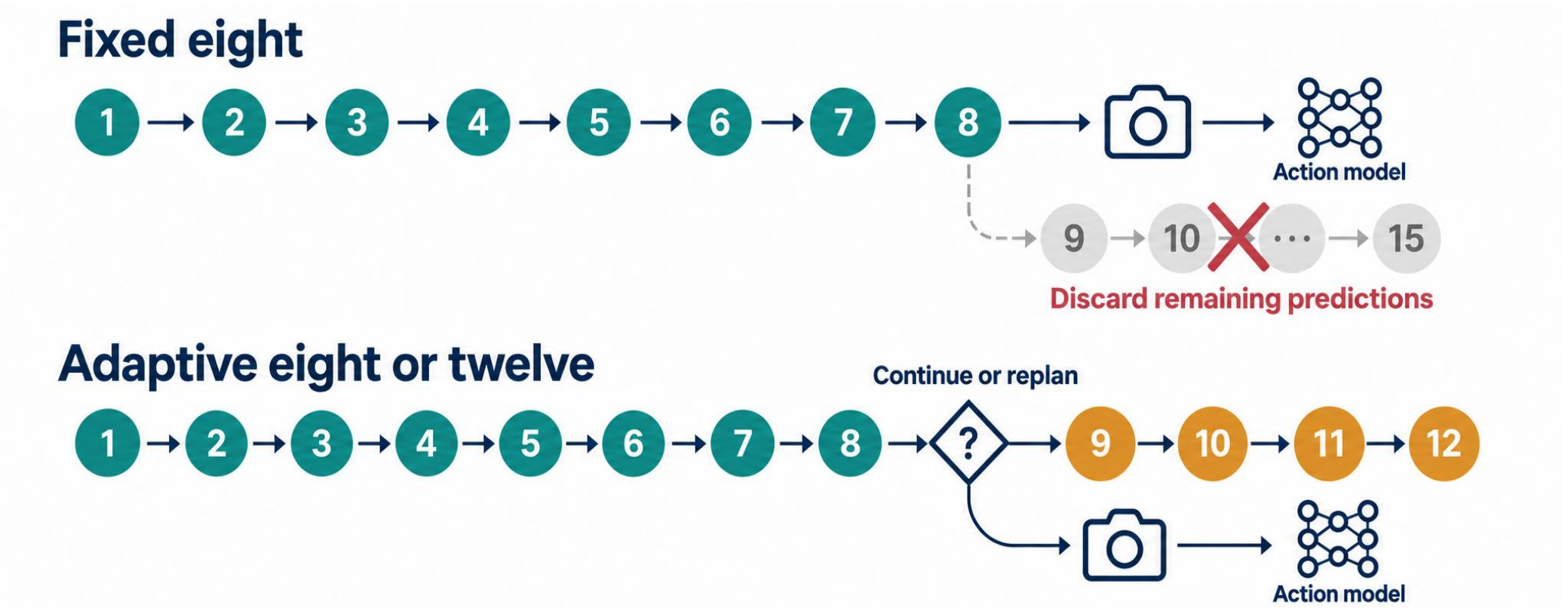
The movements change the camera image

Actions specify absolute x–y pusher targets

Success requires >95% target coverage within 300 movements. Simulator geometry supplies scoring.

Adaptive execution

After movement 8, camera and path checks decide whether to continue to 12.

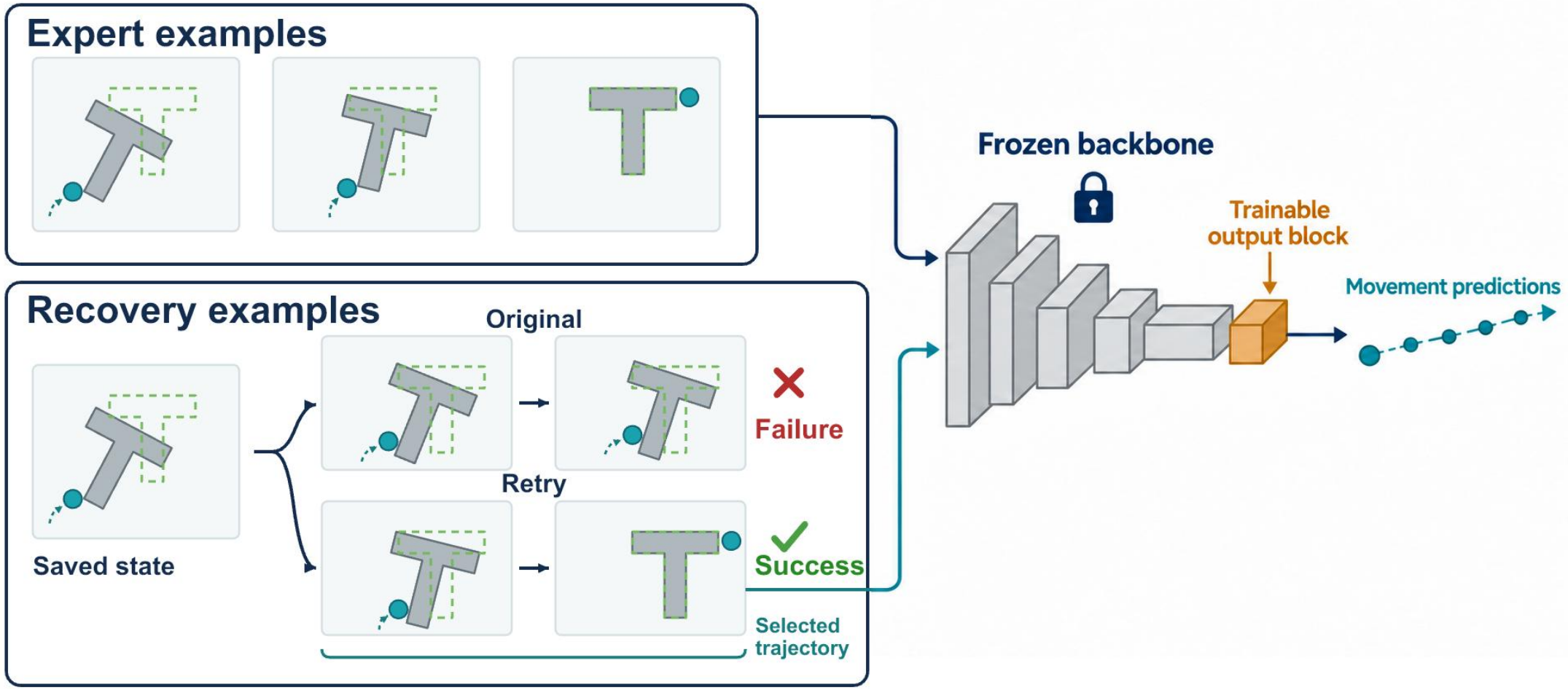


Each call generates 16 actions. The controller changes how many we execute.

Camera alignment history and the remaining path guide continuation and bounded interruptions.

Recovery fine-tuning

Mix expert demonstrations with successful retries of failed pushes.



600 updates to the final convolution block with expert rehearsal

Four matched configurations

40 starting scenes, 2 prediction draws each, 80 attempts per configuration

Configuration	Success	Alignment	Predictions	Time (s)
Original / fixed 8	57.5%	90.96%	29.03	143.26
Original / adaptive	60.0%	91.59%	25.34	135.95
Fine-tuned / fixed 8	70.0%	92.01%	27.66	136.07
Fine-tuned / adaptive	**75.0%**	**91.88%**	**22.51**	**120.92**

Adding adaptive execution to the fine-tuned model saves 18.6% of predictions

Its additional success gain is +5 percentage points, with a 95% interval of −8.75 to +18.75.

Alignment, predictions, and time are attempt averages including failures. Timing spans collection sessions.

Standard RL comparison with visual SAC

All policies use the same 40 starting scenes, observations, and commands

Policy	Successful pushes	Final alignment
Visual SAC, 100k interactions	0/40 (0%)	6.37%
Original / fixed 8	46/80 (57.5%)	90.96%
Fine-tuned / adaptive	**60/80 (75.0%)**	**91.88%**

Each Diffusion Policy configuration runs twice per scene. SAC runs once.

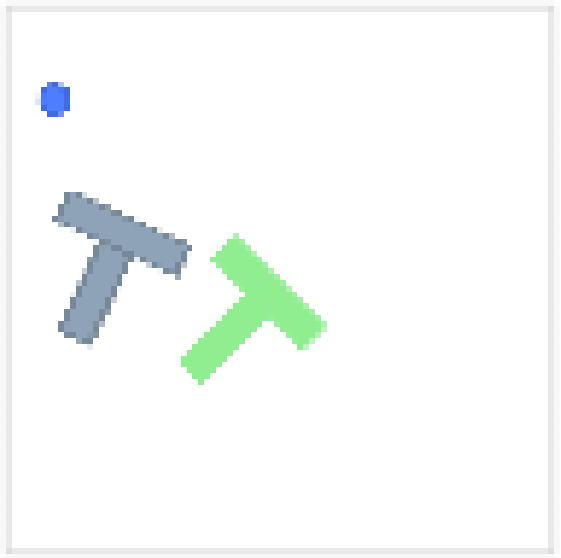
One SAC training seed, 100,000 online interactions, and about 67 minutes on this laptop

These results describe the saved checkpoint at this training budget.

A saved improvement

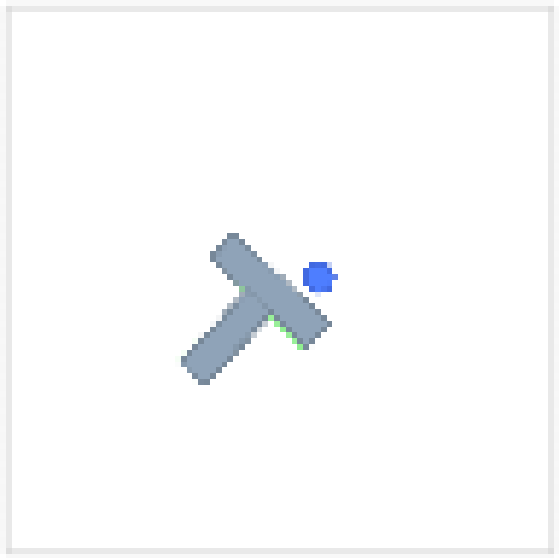
Saved scene 201, prediction draw 60000

Initial scene



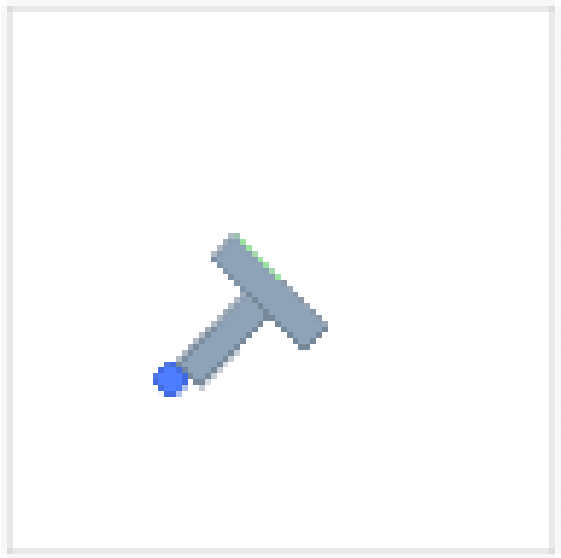
Same starting scene
and initial prediction seed

Original / fixed 8



92.21% coverage
300 movements
38 predictions

Combined method



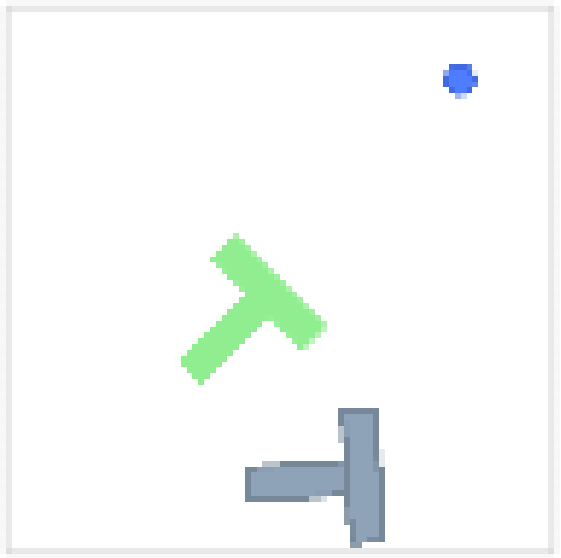
97.14% coverage
173 movements
20 predictions

Combined method means fine-tuning with adaptive execution. This is one selected example.

A lost baseline success

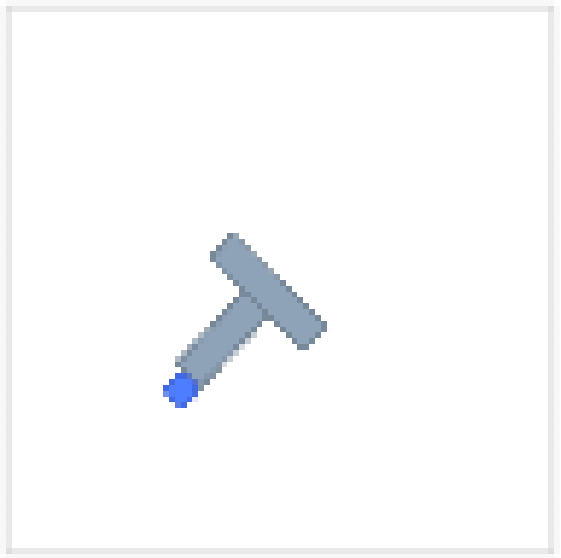
Saved scene 179, prediction draw 60000

Initial scene



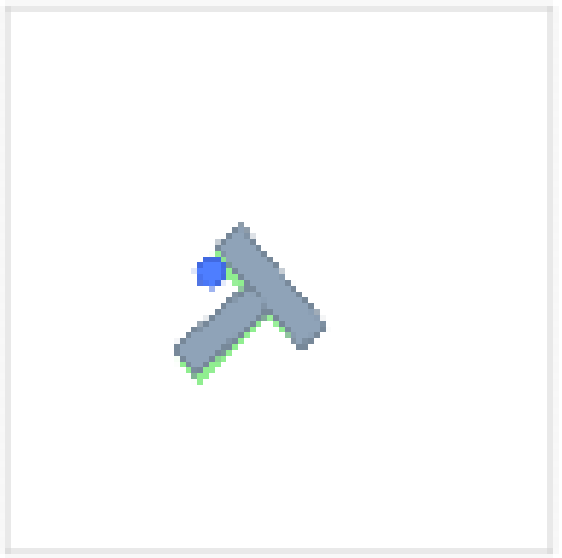
Same starting scene
and initial prediction seed

Original / fixed 8



97.53% coverage
177 movements
23 predictions

Combined method



81.78% coverage
300 movements
34 predictions

Evidence covers one task and one fine-tuning seed. Recorded runtime spans collection sessions.

Remaining work and professor feedback

Remaining work

Write the 7-page final report around the four matched configurations

Make the main results easy to reproduce from the README

Prepare the final presentation with the saved gains and regressions

Feedback for this meeting

Is the present experimental evidence sufficient for the course scope?

Should we prioritize a larger RL comparison or the remaining analysis?

Final presentation Dec. 2. Final report Dec. 13, according to the project guideline.

Thank you!



UNIVERSITY
OF ALBERTA